

IMPERIAL

Multi-Subject Factor Regression Biclustering for scRNA-seq Data

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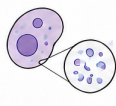
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Data & Motivation

Understanding structure in high-dimensional scRNA-seq data with pseudo-repeated measurements across subjects

What is single-cell multi-subject data?

Molecular features

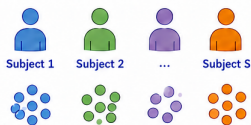


	Cell 1	Cell 2	...	Cell n
Gene 1	7	0	...	3
Gene 2	0	5	...	1
...	...	...	...	...
Gene p	2	1	...	0

Count data

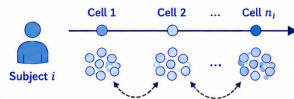
- Genes $\times$ cells matrix
- Many zeros, overdispersed

Multiple subjects (individuals)



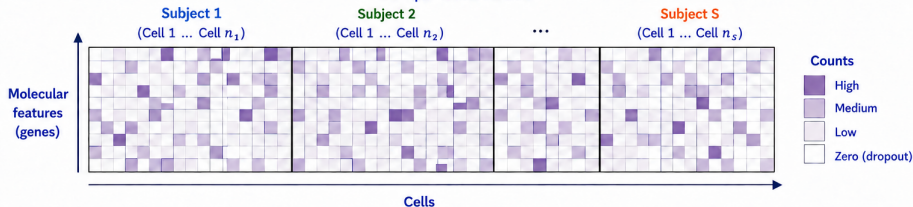
Cells obtained from different individuals
Variation across subjects (biological & technical)

Pseudo-repeated measurements within subjects



Multiple cells measured from the same individual
Cells from the same subject are more similar
(within-subject correlation)

Example data structure



How do we find patterns of co-expressed genes in subsets of cells, while accounting for dependence between cells that belong to the same individual?



Example: multi-subject scRNA-seq data

scRNA-seq data for p genes measured across three subjects, with n_1, n_2, n_3 cells.

	Subject 1				Subject 2				Subject 3			
	1	2	...	n_1	1	2	...	n_2	1	2	...	n_3
Gene 1	0	2	...	0	1	0	...	3	0	4	...	1
Gene 2	7	0	...	1	0	5	...	0	6	0	...	0
Gene 3	0	0	...	4	9	1	...	2	1	13	...	3
⋮	2	11	...	0	0	0	...	7	0	2	...	0
Gene p	⋮	⋮	⋱	⋮	⋮	⋮	⋱	⋮	⋮	⋮	⋱	⋮
	0	1	...	6	3	0	...	1	8	0	...	2

Rows = genes, columns = cells, values = expression levels

Cells exhibit within-subject dependence, excess zeros, and overdispersion

Problem set up

Aim

- Identify structured patterns and underlying biological mechanisms, i.e. find groups of subjects whose cell characteristics show similar patterns
- We cluster **subjects, not individual cells**, each subject is represented by measurements from multiple single cells

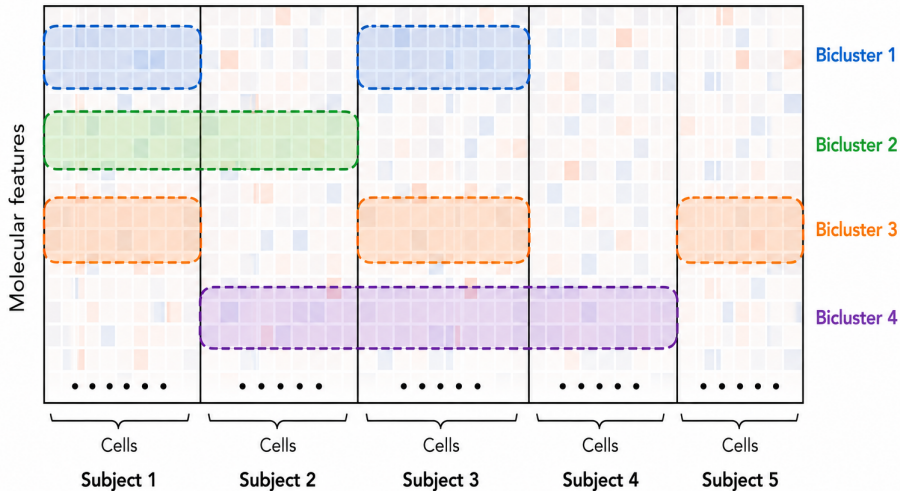
Problems

- Cells from the same individual are not independent
- Thousands of features (e.g. genes) are measured per cell
- Count data may exhibit excess zeros and overdispersion
- Covariates (e.g. age, sex, disease status) may affect the measurements
- Patterns may appear only in subsets of features and subjects

Model

A negative binomial multi-subject factor regression biclustering approach

Biclustering intuition



Sparse Bayesian factor models

Factor model

$$\mathbf{Y} = \mathbf{\Lambda}\mathbf{F} + \boldsymbol{\epsilon}, \quad \epsilon_{ji} \sim \mathcal{N}(0, \sigma_j^2),$$

with Gaussian priors, for example,

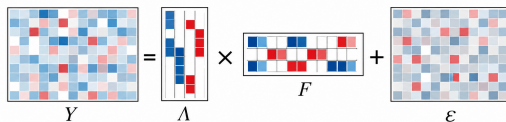
$$\lambda_{jh} \sim \mathcal{N}(0, \tau_h^2), \quad f_{hi} \sim \mathcal{N}(0, 1).$$

Sparse Bayesian factor model

Sparsity-inducing priors allow

$$\begin{aligned} \lambda_{jh} = 0 &\Rightarrow \text{feature } j \text{ is inactive in factor } h, \\ f_{hi} = 0 &\Rightarrow \text{observation } i \text{ is inactive in factor } h. \end{aligned}$$

Sparsity in both $\mathbf{\Lambda}$ and $\mathbf{F} \implies$ biclustering



Gao et al. (2016)

Biclustering via Bayesian sparse factor model

Common factor structure

$$\mathbf{Y}_s = \underbrace{\boldsymbol{\Lambda}}_{\text{feature loadings}} \underbrace{\mathbf{A}_s}_{\text{subject activation}} \underbrace{\mathbf{F}_s}_{\text{cell-level factor scores}} + \dots$$

- **Sparse feature loadings $\boldsymbol{\Lambda}$:** determine which molecular features load on each common factor
- **Subject activation $\mathbf{A}_s$:** determines whether a common factor $h \in 1, \dots, H$ is active in subject s [Grabski et al. (2023)],

$$\mathbf{A}_s = \text{diag}(\mathbf{a}_{s1}, \dots, \mathbf{a}_{sH}), \quad \mathbf{a}_{sh} \in \{0, 1\}$$

- **Cell-level factor scores $\mathbf{F}_s$:** determine how strongly an active factor is expressed in each cell. They are continuous and dense and do not determine bicluster membership

Sparse features + subject activation $\longrightarrow$ feature–subject biclusters

Accounting for within-subject dependence

Cells from the same individual share genetic and environmental backgrounds and tend to be more correlated with each other than cells from different individuals [Zimmerman et al. (2021)]. We borrow the idea from multi-study factor analysis [De Vito et al. (2019); De Vito et al. (2021)]:

$$\mathbf{Y}_s = \underbrace{\Lambda \mathbf{A}_s \mathbf{F}_s}_{\text{partially shared biclustering structure}} + \underbrace{\Upsilon_s \eta_s}_{\text{subject-specific latent structure}} + \dots$$

- Υ_s contains **subject-specific factor loadings**, accounting for effects specific to subject s .
- η_s contains the corresponding **cell-level factor scores** that reflect dependence among cells from the same subject.
- The subject-specific component therefore acts as a **latent random-effect structure**.

Partially shared factors identify biclusters, subject-specific factors account for within-subject dependence.

Accounting for observed covariates

Known biological and technical covariates may systematically affect molecular measurements and should be accounted for explicitly [De Vito and Avalos-Pacheco (2025)].

$$\mathbf{Y}_s = \underbrace{\mathbf{B}\mathbf{X}_s}_{\text{observed covariate effects}} + \underbrace{\mathbf{\Lambda}\mathbf{A}_s\mathbf{F}_s}_{\text{partially shared biclustering structure}} + \underbrace{\mathbf{\Upsilon}_s\boldsymbol{\eta}_s}_{\text{subject-specific latent structure}}$$

- $\mathbf{X}_s$ contains **observed covariates** for cells from subject s , such as age, sex, disease status, or technical variables.
- $\mathbf{B}$ contains the corresponding **feature-specific regression coefficients**.
- The regression component accounts for known sources of systematic variation, while the latent factors capture the remaining structured variation.

Regression separates known covariate effects from the latent biclustering and subject-specific structure.

Modelling single-cell count data

Single-cell RNA-seq measurements are **non-negative counts** and often exhibit substantial **overdispersion** relative to the Poisson distribution.

We therefore replace the Gaussian observation model with a **negative binomial likelihood**:

$$y_{jis} \mid r_j, \psi_{jis} \sim \text{NB} \left(r_j, \frac{\exp(\psi_{jis})}{1 + \exp(\psi_{jis})} \right),$$
$$\psi_{jis} = \beta_j^\top \mathbf{x}_{is} + \lambda_j^\top \mathbf{A}_s \mathbf{f}_{is} + \mathbf{v}_{js}^\top \boldsymbol{\eta}_{is}.$$

- r_j is a gene-specific overdispersion parameter,
- the negative binomial model allows the variance to exceed the mean:

$$\text{Var}(y_{jis} \mid -) = \text{E}(y_{jis} \mid -) + \frac{\text{E}(y_{jis} \mid -)^2}{r_j}.$$

Inference

The negative binomial likelihood is not conjugate to the Gaussian latent factor structure. Therefore:

- We introduce **Pólya–Gamma augmentation** of Polson et al. (2013) to obtain a conditionally Gaussian representation of the negative binomial likelihood [Klami (2015); Miao et al. (2020)]
- This leads to tractable **coordinate ascent variational inference (CAVI)** updates for the regression coefficients, factor loadings, factor scores and subject-activation indicators
- Gene-specific overdispersion parameters r_j are updated using an additional **compound-Poisson / CRT augmentation** [Zhou and Carin (2012); Miao et al. (2020)]
- Identification of the common and subject-specific factors is performed in a post-processing step

Results

Simulation study

Simulated data

Dataset

- $S = 6$ subjects with $n_s = 200$ cells per subject
- $p = 100$ features
- One continuous cell-level covariate and an intercept
- Gene-specific overdispersion parameters $r_j \in [5, 20]$
- Approximately 42 % zeroes in the data

Latent structure

- $H = 4$ partially shared common factors
- $H_s = 2$ subject-specific factors per subject
- Positive correlation between cells from the same subject

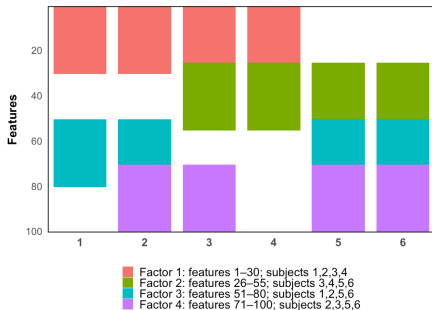
Common biclusters

- Factor 1: features 1–30, subjects 1–4
- Factor 2: features 26–55, subjects 3–6
- Factor 3: features 51–80, subjects 1, 2, 5, 6
- Factor 4: features 71–100, subjects 2, 3, 5, 6

Recovery of the bicluster structure

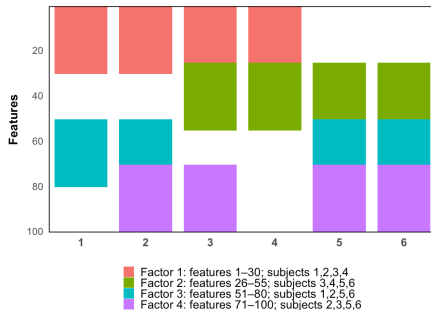
Simulated truth

Subjects

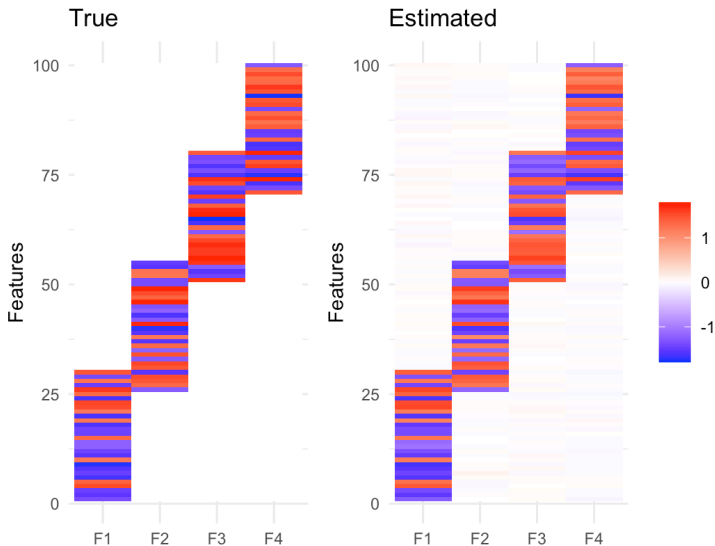


Estimated structure

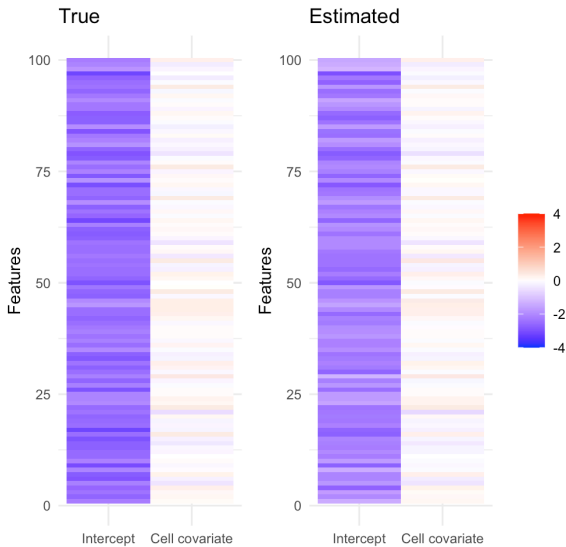
Subjects



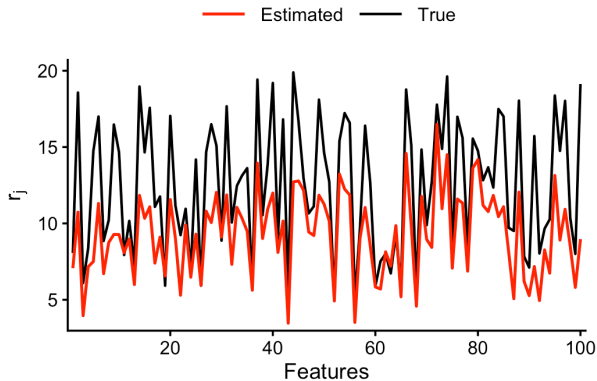
Recovery of common factor loadings



Recovery of covariate coefficients



Recovery of the overdispersion parameter



Conclusions

Multi-subject factor regression biclustering for scRNA-seq data

- Identifies sparse molecular feature patterns shared by subsets of subjects
- Accounts for within-subject cell dependence, observed covariates and overdispersion in the data
- Separates partially shared structure from subject-specific variation

Next step: clinical interpretation

- Application to single-cell data from Parkinson's disease
- Relate identified subject groups to clinical characteristics

A broader methodological question

- Few methods directly cluster subjects while accounting for within-subject single-cell variability
- We would be very interested to hear about related approaches

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